

# ESP32 Distance Measurement using Ultrasonic Sensor, I2C LCD and Blynk IoT

## Objective

To measure distance using the HC-SR04 ultrasonic sensor with an ESP32, display the measured distance on a 16x2 I2C LCD, and transmit the live readings to the Blynk IoT dashboard.

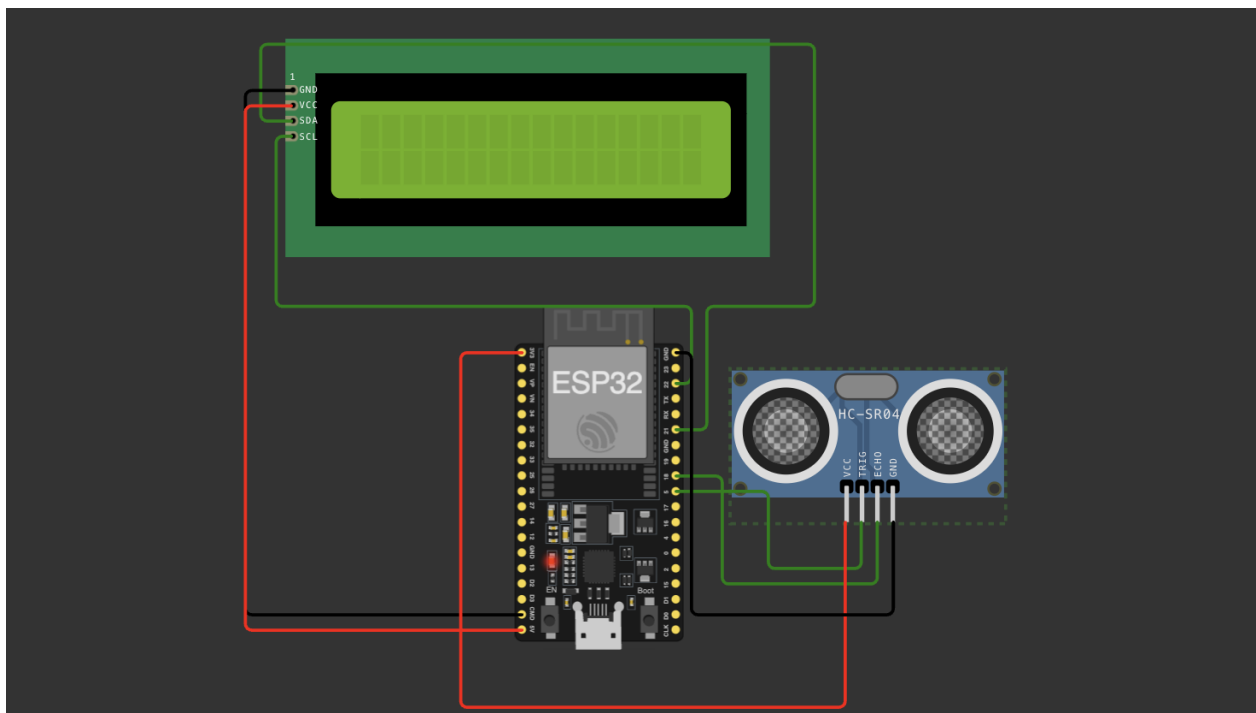
## Components Required

ESP32 Dev Board, HC-SR04 Ultrasonic Sensor, 16x2 I2C LCD, Jumper Wires, Wi-Fi Connection, Blynk IoT Platform.

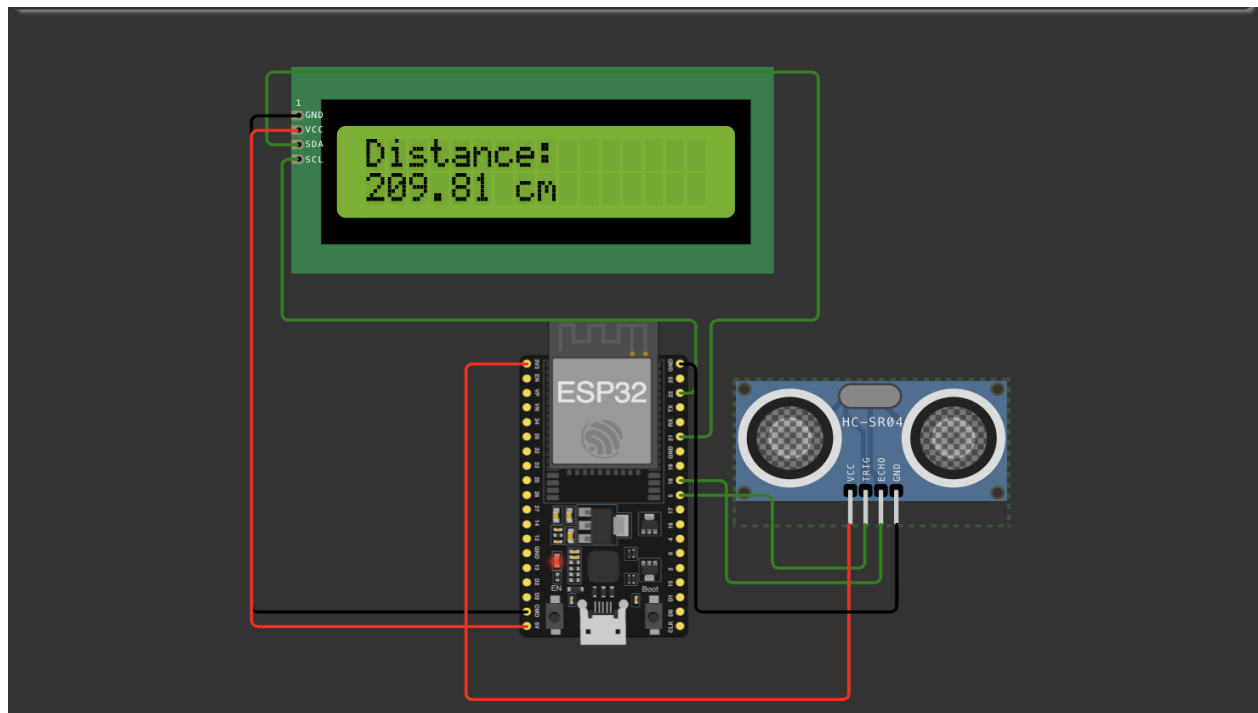
## Working Principle

The ESP32 triggers the HC-SR04 ultrasonic sensor and measures the echo pulse duration. Using the speed of sound, the distance is calculated in centimeters. The measured value is displayed on the LCD and simultaneously uploaded to the Blynk cloud using Virtual Pin V0, allowing real-time monitoring.

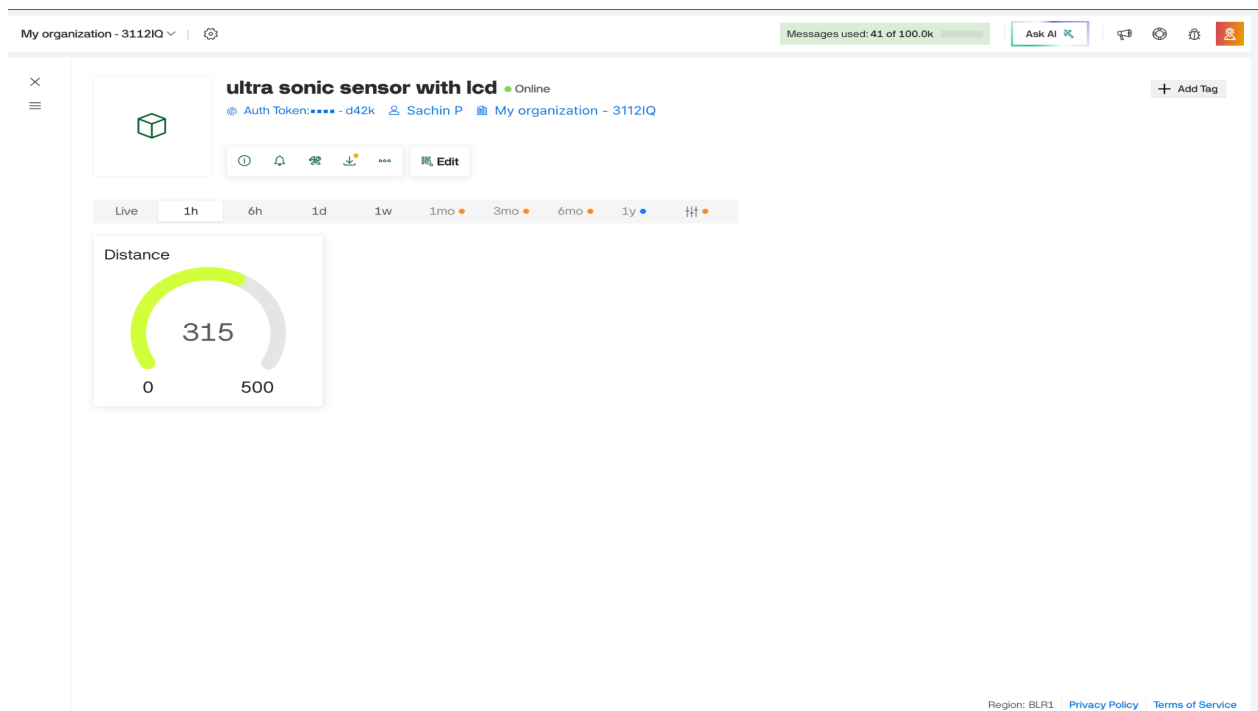
## Circuit Diagram



## Simulation Output



## Blynk Dashboard Output



## Pin Connections

ESP32 GPIO5 → HC-SR04 TRIG  
ESP32 GPIO18 → HC-SR04 ECHO  
ESP32 GPIO21 → LCD SDA  
ESP32 GPIO22 → LCD SCL  
ESP32 VIN → LCD VCC & Sensor VCC

ESP32 GND → LCD GND & Sensor GND

## Results

The circuit successfully measures the distance to nearby objects. The measured distance is displayed on the LCD in real time and updated on the Blynk dashboard gauge via Wi-Fi. Serial Monitor output also confirms accurate distance measurements.

## Conclusion

The project demonstrates successful integration of an ESP32, HC-SR04 ultrasonic sensor, I2C LCD, and Blynk IoT platform for real-time distance monitoring. It can be extended for obstacle detection, smart parking, water level monitoring, and automation applications.